Carbon-Aware Temporal Shifting

A Project Work Synopsis

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Abstract

With the rapid growth of cloud computing and large-scale data centers, energy consumption and associated carbon emissions have emerged as major sustainability challenges. Most existing workload scheduling strategies focus on performance and cost optimization while ignoring the temporal variation in grid carbon intensity. As a result, computational tasks are often executed during high-emission periods, leading to avoidable environmental impact. This project proposes a carbon-aware temporal shifting framework that intelligently schedules delay-tolerant workloads to time intervals with lower carbon intensity. The proposed system utilizes historical and real-time carbon intensity data to predict low-emission time windows and dynamically adjust task execution without violating quality of service constraints. A predictive and rule-based scheduling approach is employed to balance environmental benefits with system efficiency. The solution is implemented using cloud simulation environments and Python-based analytical models. The expected outcome is a significant reduction in carbon emissions while maintaining acceptable performance levels. This approach promotes greener cloud operations and offers practical applicability in data centers, batch processing systems, and distributed cloud infrastructures.

Keywords:

1. Carbon-Aware Computing
2. Temporal Shifting
3. Green Cloud Computing
4. Carbon Intensity
5. Sustainable Scheduling

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1. INTRODUCTION

Cloud computing has become a fundamental technology supporting modern applications such as data analytics, artificial intelligence, and large-scale enterprise systems. However, the operation of cloud data centers demands massive electrical power, leading to increased greenhouse gas emissions. As sustainability becomes a critical concern, optimizing cloud operations for environmental impact has gained significant attention.

1.1 Problem Definition

The rapid adoption of cloud computing, large-scale data centers, and distributed computing systems has significantly increased global energy consumption. Modern digital services such as data analytics, machine learning training, video streaming, and enterprise applications rely heavily on cloud infrastructure that operates continuously to meet performance demands. While cloud providers have improved hardware efficiency and adopted renewable energy sources, a substantial portion of data center power is still derived from carbon-intensive electricity grids. Additionally, traditional workload scheduling mechanisms execute tasks as soon as resources are available, without considering the environmental impact associated with the timing of energy consumption.

1.2 Problem Overview

Climate change and carbon emission reduction have become urgent global priorities, with governments, industries, and technology providers committing to net-zero and carbon-neutral goals. Cloud computing now represents a non-negligible share of global electricity usage, making it a critical area for sustainability optimization. Many cloud workloads, such as batch processing, backups, analytics, and non-real-time computations, are delay-tolerant in nature. However, executing these workloads during periods of high grid carbon intensity leads to avoidable emissions. Addressing this inefficiency is essential to reducing the environmental footprint of digital infrastructure without requiring major hardware changes or sacrificing system performance.

1.3 Hardware Specification

- Processor: Intel Core i5 / AMD Ryzen 5 or higher
- RAM: Minimum 8 GB
- Storage: 256 GB SSD or above
- Network: Stable broadband internet connection

1.4 Software Specification

- Programming Language: Python
- Operating System: Windows / Linux
- Simulation Tools: CloudSim / Custom Scheduler
- Libraries: NumPy, Pandas, Matplotlib
- Development Environment: Jupyter Notebook / VS Code

1.5 Industrial and Technological Significance

From an industry perspective, cloud service providers are under increasing pressure to meet environmental regulations, reduce operational emissions, and demonstrate sustainable computing practices. Societally, users and organizations are becoming more conscious of the environmental impact of digital services, driving demand for greener technology solutions. Technologically, the availability of real-time and historical carbon intensity data, combined with advancements in predictive analytics and intelligent scheduling algorithms, enables the development of carbon-aware computing systems. Carbon-aware temporal shifting leverages these advancements by aligning computational demand with periods of cleaner energy availability, making it a practical and scalable approach to sustainable cloud computing.

2. LITERATURE SURVEY

Recent research in sustainable cloud computing focuses on reducing data center carbon emissions by integrating environmental metrics into workload scheduling. Carbon-aware computing techniques adjust task execution based on energy sources, grid carbon intensity, and workload flexibility. Studies show that aligning computational demand with cleaner energy periods can significantly reduce emissions without major infrastructure changes. However, existing approaches such as geographical and temporal load shifting face challenges related to scalability, data locality, and real-world deployment.

2.1 Existing System

In existing cloud environments, workload scheduling primarily optimizes performance, cost, and resource utilization, executing tasks immediately without considering grid carbon intensity. Although renewable energy is used by some providers, variations in the energy mix cause periods of high emissions. Current carbon-aware methods often depend on geographical workload migration, leading to increased latency, higher data transfer costs, and data locality issues. Moreover, the lack of predictive scheduling and limited use of real-time carbon intensity data reduce the effectiveness of emission reduction for delay-tolerant workloads.

2.2 Proposed System

The proposed system presents a carbon-aware temporal shifting framework that reduces emissions by scheduling delay-tolerant cloud workloads during low carbon-intensity periods. It utilizes historical and real-time carbon intensity data to predict low-emission time windows and adjust task execution dynamically. Workloads are classified based on urgency, allowing time-sensitive tasks to run immediately while non-urgent tasks are deferred within quality-of-service constraints. By using predictive analytics and adaptive scheduling, the system balances sustainability with performance, avoids geographical migration, and offers a scalable solution for greener cloud computing.

2.3 Literature Review Summary **(Minimum 7 articles should refer)**

Year and Citation	Article/ Author	Tools/ Software	Technique	Source	Evaluation Parameter

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3. PROBLEM FORMULATION

Cloud data centers execute a large number of computational workloads continuously to support modern digital services. These workloads vary in urgency, execution time, and flexibility. At the same time, the electricity used to power cloud infrastructure is supplied by power grids whose carbon intensity fluctuates over time depending on factors such as renewable energy availability, demand, and generation mix. Despite this temporal variation, traditional cloud workload schedulers execute tasks as soon as resources become available, without considering the carbon intensity of the electricity consumed at the time of execution.

Let the set of incoming cloud workloads be represented as $\mathbf{W} = \{\mathbf{w_1}, \mathbf{w_2}, \ldots, \mathbf{w_n}\}$, where each workload w_i is characterized by parameters such as execution time, deadline, resource requirement, and flexibility. Let $\mathbf{C(t)}$ denote the carbon intensity of the power grid at time t, expressed as carbon emissions per unit of energy consumed. The primary objective is to determine an optimal execution time t_i for each workload w_i such that total carbon emissions are minimized.

However, not all workloads can be delayed arbitrarily. Time-critical tasks must be executed immediately, while delay-tolerant workloads can be deferred within predefined deadlines. Therefore, the problem involves classifying workloads based on urgency and determining whether temporal shifting is feasible without violating quality of service constraints such as deadlines, latency, and system throughput.

The problem can thus be formulated as a constrained optimization problem where the goal is to minimize the total carbon emissions associated with workload execution, subject to constraints on task deadlines, resource availability, and performance requirements. Existing approaches often rely on geographical workload migration, which introduces latency, data locality, and operational complexity. The challenge addressed in this project is to design a carbon-aware temporal scheduling mechanism that uses historical and real-time carbon intensity data to predict low-emission time windows and schedule delay-tolerant workloads accordingly within the same data center.

In summary, the problem is to intelligently align cloud workload execution with periods of lower grid carbon intensity while ensuring system performance, reliability, and service quality are not compromised. Solving this problem enables a practical and scalable approach to reducing the environmental impact of cloud computing without requiring changes to existing hardware infrastructure.

3. OBJECTIVES

- To study existing cloud workload scheduling mechanisms and identify their limitations with respect to carbon efficiency.

- To analyze temporal variations in grid carbon intensity and their impact on cloud energy consumption.

- To design a carbon-aware temporal shifting framework for delay-tolerant cloud workloads.

- To classify cloud workloads based on urgency, execution deadlines, and flexibility constraints.

- To develop a predictive model for forecasting low carbon-intensity time windows using historical and real-time data.

- To integrate carbon awareness into scheduling decisions while maintaining quality of service (QoS) requirements.

- To implement the proposed carbon-aware scheduling strategy using cloud simulation tools and Python-based models.

- To compare the proposed approach with traditional scheduling techniques in terms of carbon emissions and performance.

- To evaluate the scalability and practicality of the proposed system under varying workload and carbon intensity conditions.

- To assess the impact of carbon-aware temporal shifting on energy efficiency and sustainability metrics in cloud computing environments.

4. METHODOLOGY

The proposed methodology adopts a structured and modular approach to implement carbon-aware temporal shifting for cloud workloads while ensuring compliance with performance and quality of service requirements.

Initially, an in-depth analysis of existing cloud workload scheduling techniques is performed to identify their limitations with respect to carbon efficiency and emission awareness. This study establishes the baseline for comparison and supports the need for carbon-aware scheduling.

Subsequently, historical and real-time grid carbon intensity data is collected and analysed to capture temporal variations in emission levels. This analysis enables the identification of high- and low-carbon time windows that are suitable for workload execution.

Cloud workloads are then modeled and classified into time-sensitive and delay-tolerant categories based on parameters such as execution deadlines, resource demands, and service-level constraints. This classification determines the feasibility of temporal shifting for each workload.

A predictive mechanism is developed to forecast short-term carbon intensity trends using data-driven analytical techniques. Based on these predictions, a carbon-aware temporal scheduling framework is designed to dynamically schedule delay-tolerant workloads during low carbon-intensity periods while ensuring immediate execution of critical tasks.

The proposed scheduling framework is implemented using cloud simulation tools and Python-based analytical models to replicate real-world cloud environments. Finally, the system is evaluated against conventional scheduling approaches using metrics such as carbon emission reduction, energy efficiency, task completion time, deadline adherence, and system throughput.

6.EXPERIMENTAL SETUP

This section describes the experimental environment, datasets, workload configuration, and evaluation methodology used to assess the effectiveness of the proposed carbon-aware temporal shifting approach.

Experimental Environment

The experiments are conducted using a simulated cloud computing environment that models a data center with configurable compute resources, virtual machines, and workload schedulers. Simulation is chosen to enable controlled experimentation under varying workload intensities and grid carbon intensity conditions. All scheduling logic, predictive components, and evaluation modules are implemented using Python-based analytical models. The simulation environment enables repeatable experiments and fair comparison between different scheduling strategies.

Workload Configuration

The workload model represents a mix of real-world cloud applications and is designed to capture heterogeneity in execution behavior. Workloads are characterized by parameters including arrival time, execution duration, resource demand, and deadline constraints. Tasks are categorized into **time-sensitive workloads**, which require immediate execution, and **delay-tolerant workloads**, which can be deferred within predefined deadlines. This classification reflects practical cloud scenarios such as interactive services and batch processing jobs.

Carbon Intensity Data Modeling

Temporal variations in grid carbon intensity are modeled using historical and near-real-time carbon intensity data. The data represents changes in the electricity generation mix over time and is discretized into time slots corresponding to scheduling intervals. These carbon intensity values serve as an external signal to guide scheduling decisions and enable the identification of low-emission execution windows.

Scheduling Strategies

To evaluate the proposed approach, two scheduling strategies are implemented:

- **Baseline Scheduling**: A conventional scheduler that assigns workloads immediately based on resource availability without considering carbon intensity.
- **Carbon-Aware Temporal Scheduling**: The proposed method, which leverages predicted carbon intensity trends to temporally shift delay-tolerant workloads to low-emission periods while respecting quality of service constraints.

Both strategies are evaluated under identical workload and resource configurations to ensure fairness.

Predictive Mechanism

A lightweight prediction model is employed to forecast short-term carbon intensity trends using historical data. The predicted values are used by the scheduler to determine feasible low-carbon time windows for workload execution. The prediction horizon is selected to balance forecasting accuracy and scheduling flexibility.

Evaluation Metrics

The performance of the proposed approach is evaluated using the following metrics:

- Total carbon emissions associated with workload execution
- Percentage reduction in emissions compared to baseline scheduling
- Average task completion time
- Deadline violation rate for delay-tolerant workloads
- System throughput and resource utilization

These metrics jointly capture environmental impact and system performance.

Experimental Procedure

Each experiment is conducted by executing the same workload set under both baseline and carbon-aware scheduling strategies. Multiple runs are performed under varying workload arrival rates and carbon intensity patterns to ensure robustness. Results are averaged across runs to minimize stochastic effects.

Reproducibility and Validity

All experimental parameters, workload characteristics, and evaluation metrics are kept consistent across experiments. This ensures reproducibility and allows an unbiased assessment of the impact of carbon-aware temporal shifting on emission reduction and system performance.

7.CONCLUSION

This paper investigated the effectiveness of carbon-aware temporal shifting as a strategy for reducing carbon emissions in cloud computing environments. By incorporating temporal variations in grid carbon intensity into workload scheduling decisions, the proposed approach enables delay-tolerant workloads to be executed during periods of lower emissions while preserving quality-of-service constraints. The experimental evaluation demonstrated that integrating carbon intensity awareness into scheduling mechanisms can lead to a measurable reduction in total carbon emissions without significantly impacting system performance metrics such as task completion time, throughput, and deadline adherence.

Unlike conventional schedulers that prioritize immediate execution, the proposed method leverages predictive carbon intensity information to balance environmental sustainability with operational efficiency. The results highlight that temporal workload shifting provides a practical alternative to geographical migration, avoiding additional latency, data locality issues, and operational complexity. Overall, the findings confirm that carbon-aware temporal scheduling is a viable and scalable approach for supporting greener cloud operations. This work contributes toward sustainable cloud resource management and lays the foundation for future research on emission-aware scheduling techniques in large-scale distributed systems.

8. TENTATIVE CHAPTER PLAN FOR THE PROPOSED WORK

CHAPTER 1: INTRODUCTION

CHAPTER 2: LITERATURE REVIEW

CHAPTER 3: OBJECTIVE

CHAPTER 4: METHODOLOGIES

CHAPTER 5: EXPERIMENTAL SETUP

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

REFERENCES

[1]